VLAA-GUI: Knowing When to Stop, Recover, and Search, A Modular Framework for GUI Automation

Qijun Han, Haoqin Tu, Zijun Wang, Haoyue Dai, Yiyang Zhou, Nancy Lau, Alvaro A. Cardenas, Yuhui Xu, Ran Xu, Caiming Xiong, Zeyu Zheng, Huaxiu Yao, Yuyin Zhou, Cihang Xie

Paper version: arXiv 2604.21375v3 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *agents prematurely declare success without verifiable evidence*

Completing planned steps does not guarantee the request is met, and loop checks are only **local**.

Motivation: three open gaps

1 **Early stopping.** Success claimed without UI evidence.

2 **Repeated loops.** Detection is local, no escalation.

3 **Unfamiliar apps.** The screen may not show the workflow.

Evaluation setup

OSWorld-Verified · 361 | WindowsAgentArena · 154 | Opus 4.6 / 4.5 Manager | Sonnet 4.6 · Gemini 3 | 15 / 50 / 100 steps | no seeds reported

- **All online:** Ubuntu (OSWorld), Windows (WAA); no offline accuracy.
- **Second platform:** WAA; module studies use a fixed 60-task subset.

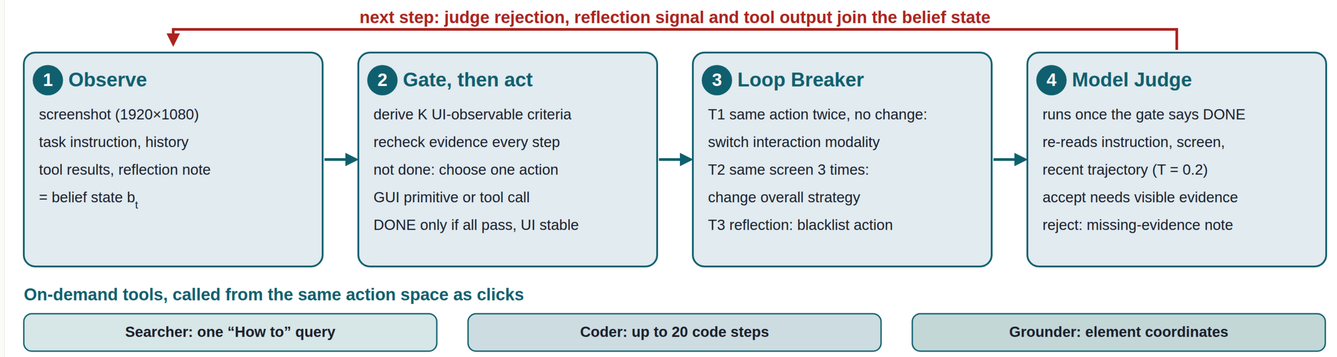
Our read (CUA Radar)

73	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; WAA; same-backbone control; step budgets; ablation; false-DONE check. **Not recorded:** seeds, cost, code.
- **Flags:** a module can hurt (Flash, 50 steps: 66.00 without CAP, 63.14 with).
- Pattern C18. Authors' numbers, not rerun.

Method: a prompted Manager with two mandatory guards and three on-demand tools

- **Sees / acts:** screenshots in; click, type, scroll, hotkey or a tool call out.
- **Mechanism:** a criteria gate plus a model judge decide DONE; a three-tier loop breaker forces a change of approach.
- **Training:** none; frontier models are prompted (prompts in the appendix).



Our redraw (not the paper's figure): one Manager step with its two mandatory guards, and the on-demand tools.

$$n_t^a \geq \tau_a = 2 \quad \text{(same action, no visible change)}, \qquad n_t^o \geq \tau_o = 3 \quad \text{(same screen)}$$

Loop Breaker triggers: fixed thresholds, same on both benchmarks

Key results: OSWorld-Verified success rate (%)

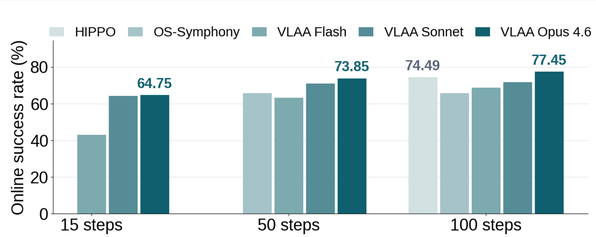


Table 1, redrawn: online, 361 tasks; baselines are HIPPO (Opus 4.5) and OS-Symphony (GPT-5).

77.45 at 100 steps (Opus 4.6); human reference 72.4

Ablation: Sonnet 4.6, 100 steps (%)

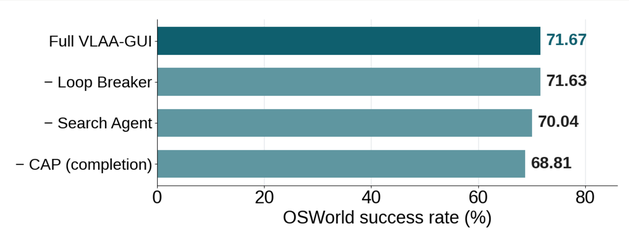


Table 2, redrawn: OSWorld, one module removed at a time.

SO WHAT → *Paper: CAP −2.86, Search −1.63, Loop Breaker only −0.04. CAP = Completeness-Aware Planner. Effects depend on backbone and step budget (Gemini 3 Flash differs).*

Headline Numbers

77.45%
OSWorld, Opus 4.6
100 steps; human reference 72.4%

61.0%	WAA, Flash
71.67%	Sonnet 4.6, OSWorld
70.34%	CoAct-1 + modules

The paper's own controls

CONTROL (SUCCESS %)	WITH MODULE(S)	COMPARED WITH
Sonnet 4.6, 100 steps	71.67 (full)	69.19 base scaffold
Flash, 50 st., CAP	63.14 with	66.00 without
Flash, 15 st., Loop Br.	43.15 with	49.30 without
WAA, 100 steps	61.0	Agent S3: 56.6

Gains depend on backbone and budget.

Takeaway

Stop, recover, search around a prompted Manager: 77.45% on OSWorld-Verified; gains vary by model and budget.

proprietary backbones | hand-set thresholds | false DONE persists

Scan to Read

Paper | Code